

LAB PROGRAM – TOPIC 5

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**SIMATS Online Programming Lab**

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LAB PROGRAM - TOPIC 5

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Language: Python C C++ Java

QUESTIONS

Q1

Search an Element in a Sorted Array

10 marks

Q2

Find the Index of a Given Number

10 marks

Q3

Count the Number of Iterations in Binary Search

10 marks

Q4

Find the First Occurrence of an Element

10 marks

Q5

Find the Last Occurrence of an Element

10 marks

Q6

Highest and Lowest Temperature Recorded

10 marks

Q7

Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Largest and Smallest Building

10 marks

Q1 Search an Element in a Sorted Array

10 marks

Given a sorted array of integers and a key value, use the Binary Search (Divide and Conquer) method to determine whether the key is present in the array. If found, print its position; otherwise, print "Element not found".

Your Code

```
1 def binary_search(arr, key):
2     low = 0
3     high = len(arr) - 1
4     while low <= high:
5         mid = (low + high) // 2
6         if arr[mid] == key:
7             return mid
8         elif arr[mid] < key:
9             low = mid + 1
10        else:
11            high = mid - 1
12    return -1
13
14 def main():
15     try:
16         arr = list(map(int, input().split()))
17         key = int(input())
18         result = binary_search(arr, key)
19         if result != -1:
20             print(result)
21         else:
22             print("Element not found")
23     except ValueError:
24         pass
25 if __name__ == "__main__":
26     main()
```

Submitted

Input

10 20 30 40 50
30

Output

2

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Highest and Lowest Temperature Recorded

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Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Largest and Smallest Building

10 marks

Q2 Find the Index of a Given Number

10 marks

Given a sorted array and a target number, find the index of the target using Binary Search. Assume array indexing starts from 0.

Your Code

```
1 def binary_search_index(arr, target):
2     low, high = 0, len(arr) - 1
3
4     while low <= high:
5         mid = (low + high) // 2
6
7         if arr[mid] == target:
8             return mid
9         elif arr[mid] < target:
10            low = mid + 1
11        else:
12            high = mid - 1
13
14    return -1
15
16 arr = list(map(int, input().split()))
17 target = int(input())
18
19 print(binary_search_index(arr, target))
```

Submitted

Input

10 20 30 40 50
4

Output

-1



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Count the Number of Iterations in Binary Search

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Find the Last Occurrence of an Element

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Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Largest and Smallest Building

10 marks

Q3 Count the Number of Iterations in Binary Search

10 marks

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Your Code

```
1 def binary_search_with_iterations(arr, key):
2     low = 0
3     high = len(arr) - 1
4     iterations = 0
5
6     while low <= high:
7         iterations += 1
8         mid = (low + high) // 2
9
10        if arr[mid] == key:
11            return mid, iterations
12        elif arr[mid] < key:
13            low = mid + 1
14        else:
15            high = mid - 1
16
17    return -1, iterations
18
19
20 try:
21
22     arr = list(map(int, input().split()))
23
24     key = int(input())
25
26     index, count = binary_search_with_iterations(arr, key)
27
28     if index != -1:
29         print(f"Element found at index: {index}")
30     else:
31         print("Element not found")
32
33     print(f"Number of iterations: {count}")
34
35 except EOFError:
36     pass
```

Input

2 3 4 5 6 7 8 9

4

Output

Element found at index: 2

Number of iterations: 3



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Q7

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10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Largest and Smallest Building

10 marks

Q4 Find the First Occurrence of an Element

10 marks

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Your Code

```
1 def find_first_occurrence(arr, key):
2     low = 0
3     high = len(arr) - 1
4     result = -1
5
6     while low <= high:
7         mid = (low + high) // 2
8
9         if arr[mid] == key:
10            result = mid
11            high = mid - 1
12        elif arr[mid] < key:
13            low = mid + 1
14        else:
15            high = mid - 1
16
17    return result
18
19
20 if __name__ == "__main__":
21     try:
22
23         data = input().split()
24         if not data:
25             arr = []
26         else:
27             arr = [int(x) for x in data]
28
29         key = int(input())
30
31         print(find_first_occurrence(arr, key))
32     except EOFError:
33         pass
```

Input

12 23 45 67 12 5 6

5

Output

-1

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Q5

Find the Last Occurrence of an Element

10 marks

Q6

Highest and Lowest Temperature Recorded

10 marks

Q7

Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q5 Find the Last Occurrence of an Element

10 marks

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Your Code

```
1 def last_occurrence(arr, key):
2     low, high = 0, len(arr) - 1
3     ans = -1
4
5     while low <= high:
6         mid = (low + high) // 2
7
8         if arr[mid] == key:
9             ans = mid
10            low = mid + 1
11        elif arr[mid] < key:
12            low = mid + 1
13        else:
14            high = mid - 1
15
16    return ans
```

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Input

1 2 2 2 3 4
2

Output

3



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QUESTIONS

Q1

Search an Element in a Sorted Array

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Find the Index of a Given Number

10 marks

Q3

Count the Number of Iterations in Binary Search

10 marks

Q4

Find the First Occurrence of an Element

10 marks

Q5

Find the Last Occurrence of an Element

10 marks

Q6

Highest and Lowest Temperature Recorded

10 marks

Q7

Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Fastest and Slowest Race Times

10 marks

Q6 Highest and Lowest Temperature Recorded

10 marks

A weather station records temperatures over a week. Find the minimum and maximum temperature using the Divide and Conquer approach.

Your Code

```
1 def find_min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4     if high == low + 1:
5         if arr[low] < arr[high]:
6             return arr[low], arr[high]
7         else:
8             return arr[high], arr[low]
9     mid = (low + high) // 2
10    left_min, left_max = find_min_max(arr, low, mid)
11    right_min, right_max = find_min_max(arr, mid + 1, high)
12    overall_min = min(left_min, right_min)
13    overall_max = max(left_max, right_max)
14    return overall_min, overall_max
15
16 def main():
17     try:
18         input_data = input().split()
19         if not input_data:
20             return
21         temperatures = [float(x) for x in input_data]
22
23         if len(temperatures) == 0:
24             return
25         min_temp, max_temp = find_min_max(temperatures, 0, len(temperatures) - 1)
26         print(f"Minimum Temperature: {min_temp}")
27         print(f"Maximum Temperature: {max_temp}")
28     except EOFError:
29         pass
30     except ValueError:
31         pass
32
33 if __name__ == "__main__":
34     main()
```

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Input

23 25.5 19 28 21 24

Output

Minimum Temperature: 19.0
Maximum Temperature: 28.0

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Find the Index of a Given Number
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10 marks**Q5**
Find the Last Occurrence of an Element
10 marks**Q6**
Highest and Lowest Temperature Recorded
10 marks**Q7**
Best and Worst Student Marks
10 marks**Q8**
Stock Price Analysis
10 marks**Q9**
Fastest and Slowest Race Times
10 marks**Q10**
Largest and Smallest Building Heights
10 marks**Q7** Best and Worst Student Marks

10 marks

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Your Code

```
1 def find_max_min(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4     if high == low + 1:
5         if arr[low] < arr[high]:
6             return arr[high], arr[low]
7         else:
8             return arr[low], arr[high]
9     mid = (low + high) // 2
10    max1, min1 = find_max_min(arr, low, mid)
11    max2, min2 = find_max_min(arr, mid + 1, high)
12    overall_max = max(max1, max2)
13    overall_min = min(min1, min2)
14
15    return overall_max, overall_min
16
17 def main():
18     try:
19         data = input().split()
20         if not data:
21             return
22
23         marks = [int(x) for x in data]
24
25         if len(marks) == 0:
26             return
27
28         max_val, min_val = find_max_min(marks, 0, len(marks) - 1)
29         print(f"Highest: {max_val}")
30         print(f"Lowest: {min_val}")
31     except EOFError:
32         pass
33
34 if __name__ == "__main__":
35     main()
```

Input

10 20 5 30 25

Output

Highest: 30
Lowest: 5

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10 marks**Q5**
Find the Last Occurrence of an Element
10 marks**Q6**
Highest and Lowest Temperature Recorded
10 marks**Q7**
Best and Worst Student Marks
10 marks**Q8**
Stock Price Analysis
10 marks**Q8** Stock Price Analysis

10 marks

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Your Code

```
1 def stock_min_max(prices):
2     if not prices:
3         return None, None
4
5     min_price = prices[0]
6     max_price = prices[0]
7
8     for p in prices[1:]:
9         if p < min_price:
10            min_price = p
11         if p > max_price:
12            max_price = p
13
14    return min_price, max_price
15
16 prices = list(map(int, input().split()))
17 min_price, max_price = stock_min_max(prices)
18 print(min_price, max_price)
```

Submitted

Input

10 12 7 17 9

Output

7 17



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Q4

Find the First Occurrence of an Element

10 marks

Q5

Find the Last Occurrence of an Element

10 marks

Q6

Highest and Lowest Temperature Recorded

10 marks

Q7

Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Q9 Fastest and Slowest Race Times

10 marks

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Your Code

```
1 def find_fastest_and_slowest(times):
2     if not times:
3         return None, None
4
5     fastest = min(times)
6     slowest = max(times)
7
8     return fastest, slowest
9
10 if __name__ == "__main__":
11     try:
12
13         user_input = input()
14         race_times = [float(t) for t in user_input.split()]
15
16         if race_times:
17             fastest, slowest = find_fastest_and_slowest(race_times)
18             print(f"{fastest} {slowest}")
19         else:
20             print("No times provided")
21     except ValueError:
22         print("Invalid input. Please enter numerical values.")
```

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Input

12.5 10.2 15.8 9.8 14.1

Output

9.8 15.8



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Q5

Find the Last Occurrence of an Element

10 marks

Q6

Highest and Lowest Temperature Recorded

10 marks

Q7

Best and Worst Student Marks

10 marks

Q8

Stock Price Analysis

10 marks

Q9

Fastest and Slowest Race Times

10 marks

Q10

Q10 Largest and Smallest Building Heights

10 marks

An urban planner is analyzing building heights in a city block and wants to determine the tallest and shortest buildings.

Your Code

```
1 def find_heights():
2     try:
3         heights = list(map(int, input().split()))
4
5         if not heights:
6             return
7
8         tallest = max(heights)
9         shortest = min(heights)
10        print(f"Tallest Building: {tallest}")
11        print(f"Shortest Building: {shortest}")
12
13    except ValueError:
14        print("Please enter valid integer heights.")
15
16 if __name__ == "__main__":
17     find_heights()
```

Submitted

Input

5
12 5 19 3 10

Output

Tallest Building: 19
Shortest Building: 3